

SMRTTECH 4AI3 - Artificial Intelligence and Machine Learning

Lab 4 - Feature Selection & Regression

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Due: Midnight 11:59PM, day before your next lab session

Please read and follow these instructions:

- Submit your lab on Avenue to Learn (A2L) before the due date (midnight of the day before your next lab session). The submission will be closed after the deadline.
- Do your lab exercises on your own, i.e., do not copy. If two lab reports (or some parts of them) are found to be copies of each other, then BOTH shall receive 0 mark.
- You may use Google Colab or any other Python programming platform of your own choice to run your programs.
- Please read the questions carefully and use AI / machine learning libraries only when specially required as part of the question. Also add comments in your code where ever necessary.
- Submit your working python source files (or Jupyter notebooks) in zipped or source code format.

1 Predict "MPG" of an Auto

Using the dataset Auto MPG (<https://archive.ics.uci.edu/ml/datasets/auto+mpg>) which is also given at avenue to learn, answer the below questions.

Dataset, has 9 columns as per below. Here, MPG is label/Y/Output, and all the other columns are features/variables/X/Input.

1. mpg: continuous
2. cylinders: multi-valued discrete
3. displacement: continuous
4. horsepower: continuous
5. weight: continuous
6. acceleration: continuous
7. model year: multi-valued discrete
8. origin: multi-valued discrete

9. car name: string (unique for each instance)

create a new Python script either as a notebook(.ipynb) or using IDE(.py) answer the below questions:

2 Exercises (20 Points)

1. Load the dataset using pandas.

```
url = './auto-mpg.data'

column_names = ['MPG', 'Cylinders', 'Displacement', 'Horsepower',
                'Weight', 'Acceleration', 'Model_Year', 'Origin']

dataset = pd.read_csv(url, names=column_names, na_values='?',
                      comment = '\\t', sep='\\t', skipinitialspace=True)
```

2. Visualize the data

- Display pandas' describe function output
- Display pandas' info function output
- Draw Seaborn pair plot for MPG, Cylinders, Displacement and Weight features.

3. Pre-process the data

- Column Origin is Categorical which has three values 1 – USA, 2 – EUROPE, JAPAN – 3. Using one hot encoding, split it into three columns named USA, EUROPE and Japan. (Hint: Use .map() function to map 1,2, 3 to string USA, EUROPE and JAPAN. Use pandas pd.get_dummies() to perform one hot encoding). The output should look like below:

```
dataset.head()
```

	MPG	Cylinders	Displacement	Horsepower	Weight	Acceleration	Model Year	Europe	Japan	USA
0	18.0	8	307.0	130.0	3504.0	12.0	70	0	0	1
1	15.0	8	350.0	165.0	3693.0	11.5	70	0	0	1
2	18.0	8	318.0	150.0	3436.0	11.0	70	0	0	1
3	16.0	8	304.0	150.0	3433.0	12.0	70	0	0	1
4	17.0	8	302.0	140.0	3449.0	10.5	70	0	0	1

- Clean the data (drop NA observations)
 - Verify you have no NA value in your dataset. (Hint use pandas dataframe isna().sum() function)
 - Split it into Train, Test and Validation dataset
 - Standardize the dataset (Hint: Standardize train, validation and test dataset separately)
4. Train the data using Scikit Linear regression model (LinearRegression)
5. Find the score of the model. (Use LinearRegression.score function)

6. Use only one feature as X and MPG as label answer the following questions
Note: You may need to reshape the data as `numpy.reshape(-1,1)` in order to use it as an argument to the score function.
- Using only Cylinders predict label MPG, Find score
 - Using only Displacement predict label MPG, Find score
 - Using only Horsepower predict label MPG, Find score
 - Using only Weight predict label MPG, Find score
 - Using only Acceleration predict label MPG, Find score
 - Using only Model Year predict label MPG, Find score
7. Find the best 3 features that gives best score
8. Using three best features, find the combined score.
9. Use the `SelectKBest` of `sklearn.feature_selection` to find the best 3 features and compare the result.